

Pathological aspects of lipid peroxidation.

Lipid peroxidation (LPO) product accumulation in human tissues is a major cause of tissular and cellular dysfunction that plays a major role in ageing and most age-related and oxidative stress-related diseases. The current evidence for the implication of LPO in pathological processes is discussed in this review. New data and literature review are provided evaluating the role of LPO in the pathophysiology of ageing and classically oxidative stress-linked diseases, such as neurodegenerative diseases, diabetes and atherosclerosis (the main cause of cardiovascular complications). Striking evidences implicating LPO in foetal vascular dysfunction occurring in pre-eclampsia, in renal and liver diseases, as well as their role as cause and consequence to cancer development are addressed.